Analytical Geometry also known as Co-ordinate Geometry

In this section we will be working with information on the Cartesian plane. It is often very helpful to sketch the given information if you are not given a sketch.

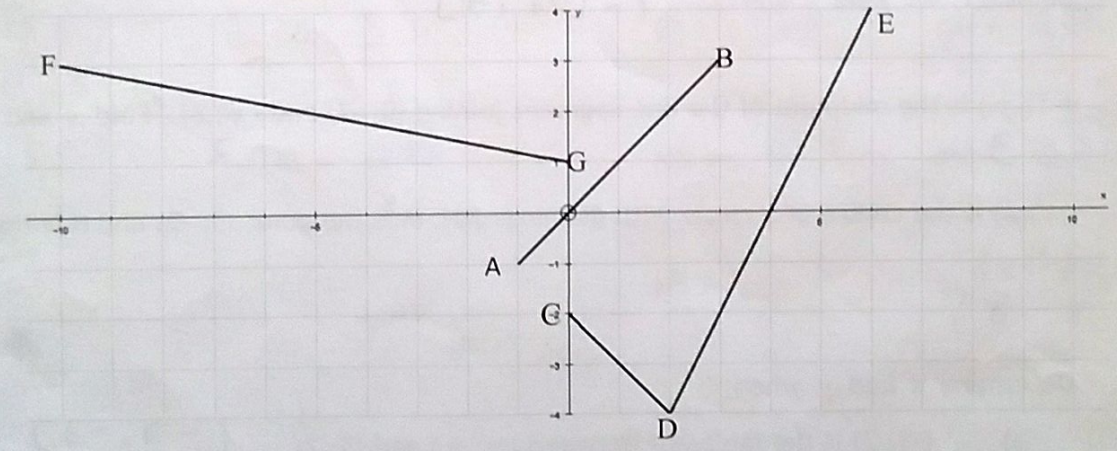
We will be working with the following:

- Midpoints of lines
- Gradient of lines
- Inclination of lines
- Equations of lines
- Length of lines (distance)
- Quadrilaterals

Some of these concepts are new and some will simply need to be revised ☺.

1. MIDPOINT

Each of the given lines has a mid-point. Give the co-ordinates of the midpoint of each line



a) Midpoint AB (1 ; 1)
Midpoint CD (1 ; -3)
Midpoint DE (4 ; 0)
Midpoint FG (-5 ; 2)

These you read off the drawing BUT there is another way to work the answers out.....

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2} \; ; \; \frac{y_1 + y_2}{2}\right)$$

Now let's practise using the formula.....

Midpoint Exercise Pg 3 "Memo"

1 a) $M\left(\frac{3+5}{2}; \frac{7-2}{2}\right) = M\left(4; 2\frac{1}{2}\right)$

b) $M\left(\frac{0-2}{2}; \frac{4-4}{2}\right) = M(-1:0)$

c) $M\left(\frac{-3-5}{2}; \frac{-2+3}{2}\right) = M\left(-4; \frac{1}{2}\right)$

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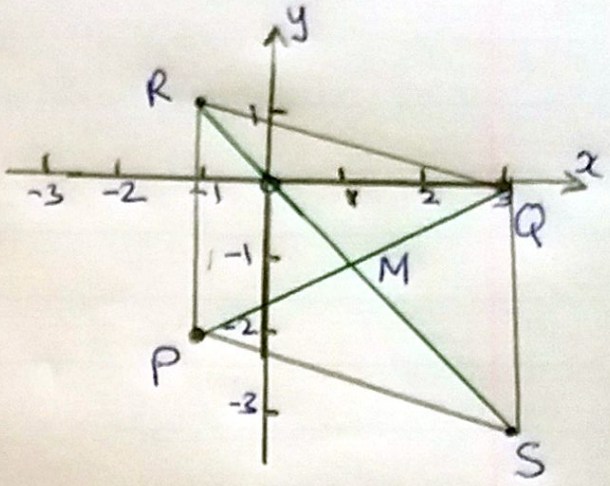
2 P(-1: -2) ; Q(3: 0) ; R(-1: 1) S(3: -3)

$M_{PQ}\left(\frac{-1+3}{2}; \frac{-2+0}{2}\right)$

$= (1 ; -1)$

$M_{RS}\left(\frac{-1+3}{2}; \frac{1-3}{2}\right)$

$= (1 ; -1)$



∴ PQ and RS have same midpoint M(1: -1) so PQ & RS bisect each other.

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3 A(-3: 2) ; B(-1: 1) ; C(3: 3) R(1: 4)

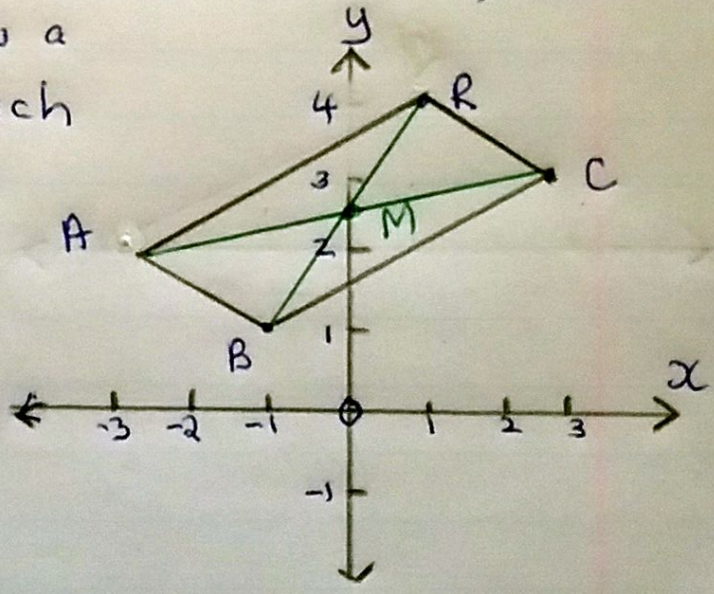
Note: you need to draw a sketch to determine which lines are the diagonals

$M_{AC}\left(\frac{-3+3}{2}; \frac{2+3}{2}\right)$

$= (0 ; 2\frac{1}{2})$

$M_{BR}\left(\frac{-1+1}{2}; \frac{1+4}{2}\right)$

$= (0: 2\frac{1}{2})$



AC and BR share midpoint M(0: 2½)

∴ Diagonals of ABCR bisect each other